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ORIGINAL ARTICLE

A simulation model of selective laser melting (SLM) of glass silica monolayer

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Abstract
Although experimental research demonstrates the feasibility of processing glass via selective laser melting (SLM), achieving SLM precision manufacturing of fused silica is still a challenging but promising field. Therefore, a numerical model was developed in this work to predict the densification kinetics of glass powder monolayer under SLM, which provided a simulation tool for the optimization design of processing parameters. To be particular, the relative neck diameter and density of the powder bed under SLM were calculated using a cluster model to simulate the degree of consolidation, from which the thermal diffusivity of the powder bed was determined and the corresponding results of heat transfer were obtained through the numerical model implemented by using finite difference method (FDM). The experimental results for the SLM of glass with different processing parameters from other references validated the simulation results with a prediction error of less than 2%. Hence, the effects of processing settings, such as the influence of particle size and preheating temperature, could be accurately investigated by using this numerical modelling method. It was found that using powders with smaller particle sizes and setting a higher preheat temperature could improve the manufacturing quality of the obtained glass parts. From the densification results of the numerical modelling, we also introduced a novel optimization criterion, which could be utilized to optimise the processing parameters and fabricate the high-purity glass parts via SLM.

Keywords Selective laser melting (SLM) · Silica glass · Simulation model · Processing optimisation

1 Introduction

Additive manufacturing (AM), or 3D printing, is a manufacturing technology in which a three-dimensional object is constructed by adding material layer by layer [1]. The AM technique in this study is selective laser melting (SLM) which melts the powder bed from bottom to upper and produces a fully dense part layer-wise [2]. Accurate and rapid fabrication is the most outstanding strength of SLM over other AM methods, allowing the precision processing of micro- and nano-powders [3]. However, even though glass–ceramic material has many desirable properties, such as low thermal expansion coefficient, low electrical conductivity and high chemical resistance, applying the technique of SLM on glass is still challenging [4]. The main impediment hindering the development of SLM for glass is the infamous difficulty in shaping glass parts, which demands high temperatures and a complex casting process [5].

Over the past few years, several experimental investigations have attempted to fabricate different glass ceramic parts using SLM. Fateri and Gebhardt [6] demonstrate SLM's feasibility for soda-lime glass powder and fabricated parts with a relative density of 93%. An experimental study on quartz glass conducted by Khmyrov et al. [7] successfully fabricates a crack-free glass monolayer using SLM. Moreover, Gan and Wong [8] successfully use SLM to build porous lithium aluminosilicate (LAS) glass parts with spodumene minerals. Although not mature enough to manufacture the ideal glass ceramic objects, these studies are essential for the development of SLM for glass materials in the future.

In addition to the experimental studies of SLM for glass, simulation is an essential method that provides a deeper understanding of this laser AM process of glass ceramics. However, only a handful of numerical investigations are available for the SLM of glass or ceramic materials. To predict the thermal distribution and melt-pool geometry, Li et al. [9] put forward

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a heat conduction model for ceramic material under SLM and compared the model's volumetric and surface heating sources. Subsequently, they establish a numerical model for the convection–diffusion phase change process during the laser melting of ceramic materials [10]. In one study, a 3D finite element model of ceramic material during SLM with concurrent shrinkage is established by Chen et al. [11]. To obtain fused silica, Protasov et al. [12] successfully calculate heat transfer in the consolidation process of silica powder bed by considering the effective thermal conductivity related to the neck growth between glass particles.

However, little modelling study to date focuses on the densification process of the glass powder bed, causing the lack of criteria to optimise the processing parameters and limiting the fabrication resolution of SLM. Specifically, the thermal diffusivity of the silica powder layer is highly affected by its densification degree, determining the effectiveness of SLM processing on glass [12]. The simulation for the densification of silica powder is also indispensable since the properties of the finally obtained glass part highly depend on its relative density [13]. Besides, research on the densification process of powder monolayer under SLM solidifies the foundation of quantitively calculating the shrinkage deformation of the glass compact. Thus, a numerical model of SLM for silica glass was established in this work to predict the densification and deformation kinetics of the glass compact.

In this work, the sintering densification kinetics of the glass powders are incorporated into our numerical model using the Cluster model [14] during SLM. Before reaching the melting point, sintering occurred between glass powders due to the high temperature. Accordingly, the area between the fully dense glass compact and the initially loose powder bed is a zone of density variation dominated by glass sintering. Additionally, plenty of research demonstrates the Cluster model's feasibility on various glass materials [15, 16]. Hence, this study established a heat conduction model with sintering effects for SLM of silica glass to predict its densification process, which is numerically implemented by using the finite difference method (FDM). Similar to the previous investigations done by other researchers [7, 9–12], we focus our numerical study on the glass monolayer in the SLM process, which is fundamental for the research of parts fabrication.

2 Modelling methodology

2.1 Modelling formulation

Selective laser melting (SLM) builds parts layer by layer, and the interaction between the laser and each powder layer is similar [7, 12]. Thus, the research object of this study was the single glass layer with laser treatment, which constructs the foundation of modelling the whole SLM process. Figure 1 demonstrates the schematic for the laser treatment of the target monolayer with the corresponding coordinate systems.

As Fig. 1 illustrates, the laser beam moves along a straight line and scans the target surface along x direction at a constant speed. The heat transfer process is the same for any plane in the x-length direction. Thus, the above 3D transient heat diffusion problem (x–y-z) could be converted into a 2D transient heat diffusion problem (y–z). The 2D transient heat diffusion model was constructed as follows.

$$\begin{aligned} &\frac{\partial^2 T}{\partial y^2}+\frac{\partial^2 T}{\partial z^2}=\frac{1}{\alpha}\frac{\partial T}{\partial t}\left(\in \Omega, t>t_0\right)\\ &T=T_a\left(\in \Gamma_1, t>t_0\right)\\ &-\lambda\frac{\partial T}{\partial z}=q_0\left(\in \Gamma_2, t>t_0\right)\\ &T_0=T_a\left(\in \Omega, t=t_0\right) \end{aligned} \tag{1}$$

where Γ_1 is the Dirichlet (first) boundary condition, Γ_2 is the Neumann (second) boundary condition and $\Gamma=\Gamma_1+\Gamma_2$ is the boundary of the whole region Ω. T Refers to the temperature, T_a is the ambient temperature and T_0 is the initial temperature at the initial time t_0.

$\alpha=\lambda/\rho C_p$ is the thermal diffusivity, in which λ, C_p and ρ represent the heat conduction coefficient, specific heat and density. Under the SLM processing, the densification of the glass influences its thermal diffusivity, which can be simulated by using the sintering model illustrated in Section 2.2. Additionally, the modelling method is demonstrated in Section 2.3 for the heat conduction coefficient λ of glass under SLM. Finally, the formulated model is numerically implemented by using the finite difference method, which is depicted in Section 2.4.

q_0 is the heat flux and is approximated by the Gaussian distribution.

$$q_0=\frac{P_0}{\pi r_0^2}\exp\left(-\frac{r^2}{r_0^2}\right) \tag{2}$$

where P_0 is the incident laser power, r_0 represents the laser beam radius and $r^2=x^2+y^2$.

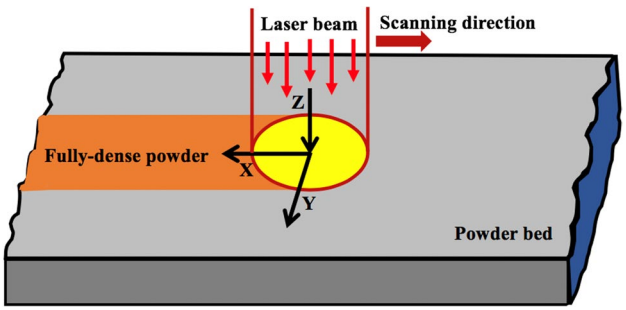


Fig. 1 Schematic for the laser treatment of the target monolayer with the corresponding coordinate systems

2.2 Consolidation of glass powders

The modelling for the glass powder consolidation was established to solve the temperature-dependent thermal diffusivity of the compact layer. In this model, densification kinetics, necking growth and porosity of the powder bed were correspondingly estimated. With the temperature increasing to a certain point, the consolidation process between glass particles behaves like the droplets of viscous liquid and tends to coalesce, as shown in Fig. 2.

The necking growth kinetics between two spherical particles of diameter D was calculated by the following [17].

$$\frac{d\xi^2}{dt} = \frac{3\delta}{\eta D} \tag{3}$$

where $\xi = a/D$ is the relative neck diameter, η denotes the dynamic viscosity, and δ represents the surface tension coefficient.

The surface tension coefficient of the silica glass was considered as an independent constant of temperature and equal to 0.4 N/m [18]. By contrast, the shear viscosity η decreases significantly with the temperature increasing. Here, the approximation equation in the previous study [12] was used to calculate the dynamic viscosity as follows.

$$\begin{cases} \eta = 3.8 \cdot 10^{-14}(\text{Pa} \cdot \text{s}) \exp\left(\frac{85{,}640\text{K}}{T}\right), & T < 1670\text{K} \\ \eta = 5.8 \cdot 10^{-8}(\text{Pa} \cdot \text{s}) \exp\left(\frac{61{,}990\text{K}}{T}\right), & T > 1670\text{K} \end{cases} \tag{4}$$

The densification kinetics of glass powders is determined by using the well-proved model: Clusters model. Before discussing the Clusters model in detail, the two basic models, the Frenkel (F) model and the Mackenzie-Shuttleworth (MS) model should be introduced first. F model provides a densification kinetics description of the early stages of sintering between spherical monodispersed particles [19]. The valid range of the F model is roughly within the first 10% of linear shrinkage. Specifically, if the initial relative density of the powder bed is 0.6, the 10% of linear shrinkage means a relative density of 0.8. The change in the relative density during sintering is estimated in the following:

$$\rho(t) = \frac{\rho_0}{\rho_g}\left(1 - \frac{3\delta t}{8\eta r}\right)^{-3} \tag{5}$$

where ρ_0 is the initial compact density, ρ_g is the glass density and $r = D/2$ denotes the initial particle radius. In contrast to the F model, the MS model explains the final stages of sintering between spherical particles with spherical pores. According to the MS model, the densification rate is given as follows:

$$\frac{d\rho}{dt} = \frac{3\delta}{2a_0\eta}(1-\rho) \tag{6}$$

in which a_0 is the initial radius of the spherical pores. The clusters model divided the compacts into several clusters of equally sized particles and carried out both Frenkel and M-S models simultaneously in these clusters. Based on the previous experiment [7], the particle size distribution is narrow enough, so the "neck forming ability" can be disregarded. For a compact with a volume fraction v_r of spherical particles of radius r, the following clusters model is valid in the whole process to determine the densification kinetics [14].

$$\rho(T) = \sum_r v_r\left[\rho_F(r,T)\theta\left(T_{0.8} - T\right) + \rho_{MS}\left(a_0(r),T\right)\theta_{MS}\left(T - T_{0.8}\right)\right] \tag{7}$$

During the early stage of sintering, the density is calculated by using the F model and thus, $\theta_F\left(t_{0.8} - t\right) = 1$ and $\theta_{MS}\left(t - t_{0.8}\right) = 0$ for $t < t_{0.8}$. When it comes to the final stage of sintering, the density is determined by using the MS model and so $\theta_{MS}\left(t - t_{0.8}\right) = 1$ and $\theta_F\left(t_{0.8} - t\right) = 0$ for $t > t_{0.8}$.

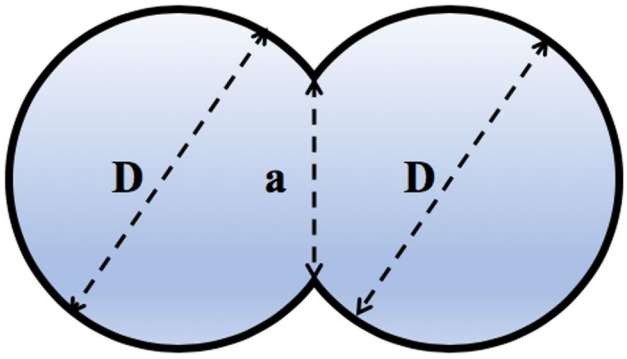


Fig. 2 Consolidation process between two spherical particles with the diameter D and the neck length a

2.3 Effective thermal conductivity

Generally speaking, heat transfer is dominated by conduction, convection and radiation. Nevertheless, the convection effect was ignored due to the micron- or even submicron-scale pore size around the glass powders of D = 20 μm. Therefore, only radiative heat transfer and conduction through the solid and gas phases were considered here.

When the neck growth is at the early stage, necks between the particles control heat conduction of the solid phase [12]. Based on the statistical network model [20], the effective thermal conductivity in the solid phase λ_{es} is given.

$$\lambda_{es} = \lambda_s \frac{(1-\varepsilon)N}{\pi}\xi \tag{8}$$

where λ_s is the thermal conductivity in the solid phase, and Eq. (8) works if $\lambda_{es} < \lambda_s$. Otherwise, λ_s is set as λ_{es}. N denotes the mean coordination number calculated from the porosity ε as [21].

$$N = \frac{\varepsilon + 3 + \sqrt{\varepsilon^2 - 10\varepsilon + 9}}{2\varepsilon} \tag{9}$$

The porosity is equal to the volume fraction of the gas phase, and the initial porosity of the compact in SLM is around 0.4. During the densification process, the porosity decreases and vanishes once the powder bed is fully dense. The relation between the porosity and the relative density ρ could be obtained as follows:

$$\varepsilon = 1 - \rho \tag{10}$$

Heat conduction through the gas phase of the compact is considerable in the tiny pores or gaps between particles [22]. In the previous study [12], the effective conductivity of the gas phase λ_{eg} is estimated with a value proportional to the conductivity of the gas λ_g.

$$\lambda_{eg} = F\lambda_g \tag{11}$$

with factor

$$F = \frac{(1-\varepsilon)N}{2}\left(\frac{1}{2}\ln\frac{1+L}{1+L\xi^2} + \ln\frac{1+\sqrt{L}}{1+\sqrt{L}\xi} + \frac{1}{1+\sqrt{L}} - \frac{1}{1+\sqrt{L}\xi}\right) \tag{12}$$

where

$$L = \frac{\gamma+1}{9\gamma-5}\cdot\frac{3}{\mathrm{Kn}4\sqrt{\pi}} \tag{13}$$

γ is the adiabatic exponent and $\mathrm{Kn} = l/D$ refers to the Knudsen number with mean free path l. The mean free path can be obtained through gas-phase conductivity λ_g using the following definition [23].

$$l = 6\frac{\gamma-1}{9\gamma-5}\frac{\lambda_g}{p}\left(\frac{mT}{2k}\right)^{1/2} \tag{14}$$

where p is the gas pressure, m denotes the molecular mass of the gas and k represents the Boltzmann constant. According to previous research [24], the thermal conductivity of air can be estimated as follows:

$$\lambda_g = 0.0244\frac{\mathrm{W}}{\mathrm{mK}}\left(\frac{\mathrm{T}}{273\mathrm{K}}\right)^{0.82} \tag{15}$$

During the process of SLM, there exists a heterogeneous medium, including a transparent phase and an opaque phase. The effective radiative thermal conductivity of the powder compact could be estimated [12].

$$\lambda_r = \frac{22.9 - 21.9p_r}{1 - \frac{N}{4}\xi^2}1.075\cdot 10^{-3}\frac{\mathrm{W}}{\mathrm{mK}}\left(\frac{\mathrm{T}}{1000\mathrm{K}}\right)^3 \tag{16}$$

where p_r is the weight of thermal radiation where the solid phase is non-transparent and p_r can be estimated by the approximation equation.

$$\frac{1}{p_r^{1/4}} = 0.66 + \frac{\mathrm{T}}{1340\mathrm{K}} \tag{17}$$

In short, the effective thermal conductivity is dependent on the neck growth and densification process of powder consolidation. Finally, we can summarise the above three components of effective thermal conductivity and obtain the total effective thermal conductivity [12].

$$\lambda = \lambda_{es} + \lambda_{eg} + \lambda_r \tag{18}$$

2.4 Numerical implementation

This paper adopted the finite difference method (FDM) to solve the transient heat conduction problem. After the uniform discretisation of the space domain, the step length along the y-axis and z-axis is $\Delta y = y_{i+1} - y_i$ and $\Delta z = z_{j+1} - z_j$, where $i, j = 0, 1, 2, 3....$ We used n to uniformly discretise the time domain and the step length was set as $\Delta t = t_{n+1} - t_n$. Thus, the term on the right of the first equation in Eq. (1) can be approximated by the first-order time forward difference method.

$$\left(\frac{\partial T}{\partial t}\right)_{i,j}^{n} = \frac{\left(T_{i,j}^{n+1} - T_{i,j}^{n}\right)}{\Delta t} + \mathrm{O}(\Delta t) \tag{19}$$

As for the second-order partial differential on the left of the same equation, we used the central difference method to calculate it.

$$\left(\frac{\partial^2 T}{\partial y^2}\right)_{i,j}^{n} = \frac{\left(T_{i+1,j}^{n} - 2T_{i,j}^{n} + T_{i-1,j}^{n}\right)}{(\Delta y)^2} + \mathrm{O}\left(\Delta y^2\right) \tag{20}$$

$$\left(\frac{\partial^2 T}{\partial z^2}\right)_{i,j}^{n} = \frac{\left(T_{i,j+1}^{n} - 2T_{i,j}^{n} + T_{i,j-1}^{n}\right)}{(\Delta z)^2} + \mathrm{O}\left(\Delta z^2\right) \tag{21}$$

After substituting Eqs. (19), (20) and (21) into (1), we obtained the difference equation of Eq. (1).

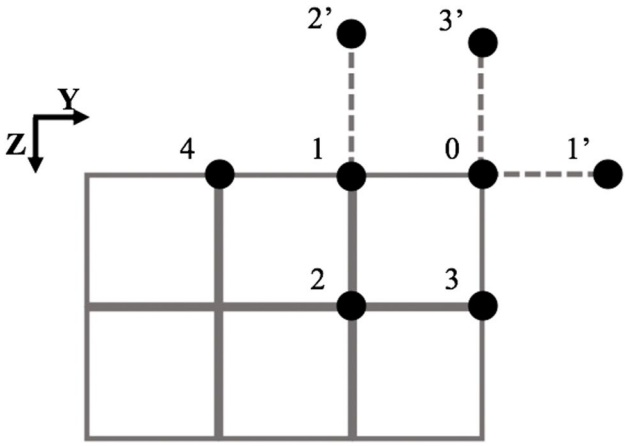


Fig. 3 Schematic of the boundary nodes, virtual nodes and interior nodes

$$\frac{T_{i+1,j}^{n}-2T_{i,j}^{n}+T_{i-1,j}^{n}}{(\Delta y)^{2}}+\frac{T_{i,j+1}^{n}-2T_{i,j}^{n}+T_{i,j-1}^{n}}{(\Delta z)^{2}}=\frac{1}{\alpha}\frac{T_{i,j}^{n+1}-T_{i,j}^{n}}{\Delta t} \tag{22}$$

Different type of boundary condition is processed differently in FDM [25]. In general, the first-type boundary node was processed as $\overline{T}\Big/2$ in the initial moment, and then the boundary node temperature remained at $\overline{T}$. $\overline{T}$ is the Dirichlet boundary condition temperature. As for the second boundary condition, the virtual node outside the boundary was set to connect the boundary node with the interior node. Take the node numbering in Fig. 3 as an instance.

The central difference method was used to replace the first-order partial differential equation (PDE) of the second boundary condition in Eq. (1).

$$-\lambda\frac{T_{2}^{n}-T_{2'}^{n}}{2\Delta z}=q_{0} \tag{23}$$

Based on the node number in Fig. 3, Eq. (23) was substituted into Eq. (21), and the temperature on node 1 could be obtained.

$$T_{1}^{n+1}=T_{1}^{n}+\alpha\Delta t\left(\frac{T_{4}^{n}-2T_{1}^{n}+T_{0}^{n}}{\Delta y^{2}}+\frac{2T_{2}^{n}-2T_{1}^{n}-\frac{2\Delta z}{\lambda}q}{\Delta z^{2}}\right) \tag{24}$$

Table 1 presents the numerical algorithm of our model.

3 Results and discussion

3.1 The result of temperature distribution

The developed numerical model was implemented by using MATLAB. The parameters used in the model are listed in Table 2.

The study plane was the y–z plane shown in Fig. 4. The numerical model was conducted with the beam diameter $2\gamma_0 = 1.5$ mm, scanning speed $u = 1$ mm/s and ambient temperature $T_a = 298$ K. Detailed laser parameters can be found in Ref [12]. In this simulation, the initial distance from the laser beam centre to the study plane was 1.5 mm. The schematic is shown in Fig. 4 for the whole process of the laser beam passing through the study plane of the glass monolayer.

The yellow part in Fig. 4 represents the domain of the laser beam followed by the orange part which refers to the consolidated glass powder. The light-blue part is the study y–z plane, of which the temperature distribution, densification degree and deformation are simulated and discussed in the following sections. The temperature distributions for

Table 2 Parameters of the modelling

Parameter	Symbol	Value	Unit
Laser beam radius	r_0	0.75	mm
Ambient temperature	T_a	298	K
Glass powder diameter	D	20	μm
Surface tension coefficient	δ	0.4	N/m
Glass density	ρ_g	2.2	g/cm^3
Initial pore radius	a_0	1	μm
Solid-phase thermal conductivity	λ_s	2.2	W/(m K)
Porosity	ε	0.4	

Table 1 Numerical implementation algorithm of the model

1	Initial conditions and initialisation
2	Determine the relative neck diameter ξ^n and the relative density ρ^n of the glass powders
3	Calculate the effective thermal conductivity λ^n and the thermal diffussity α^n
4	Discretise and calculate the 2D heat transfer model
5	Time update $t^{n+1} = t^n + \Delta t$
6	If the relative density reaches 1 or temperature starts to decrease, fix the value of the relative density as the previous value
7	Counter update and time update
8	If simulation not complete, go to 2
9	Output

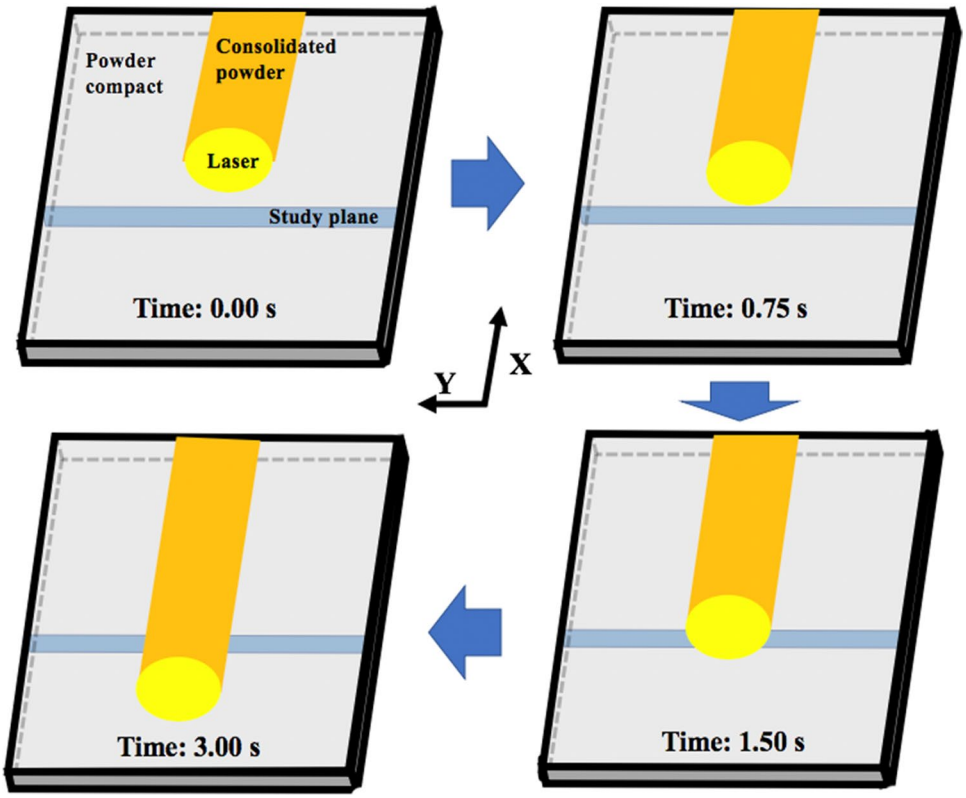


Fig. 4 The schematic for the process of the laser treatment for glass monolayer from the initial location to the final location, i.e. from 0.00 to 3.00 s

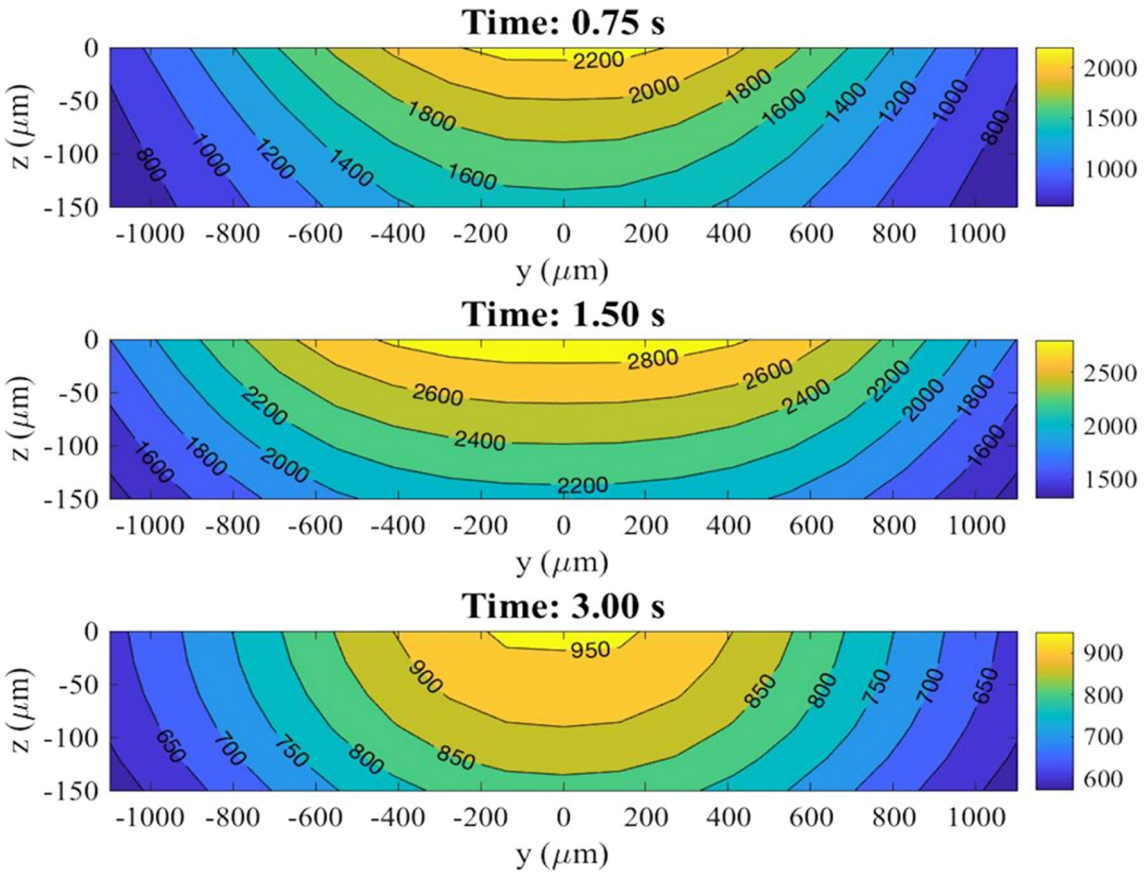


Fig. 5 Temperature distribution for the study plane of the glass monolayer during the process of SLM

Table 3 Parameters for the numerical calculations and the previous experiments

Variant	a	b	c
Laser power (W)	20	20	15
Layer thickness (μm)	150	175	175

the study plane at different times are plotted in Fig. 5 for the glass monolayer under SLM.

In Fig. 5, the simulation was implemented for laser power $P_0 = 20$ W, powder bed thickness $L = 150\mu$m and powder particle diameter $D = 20\mu$m. In the first half of time duration, the temperature tends to increase. As demonstrated in Fig. 5, the temperature of the bed surface ($z=0$) reached its peak point near the middle time and cooled down after 1.5 s.

3.2 The validation result of the densification and deformation

According to the previous experimental study from other reference papers [7, 12], the combinations of the processing parameters in this work are listed in Table 3. The experimental results in Ref. [12] under the cases of a, b and c in Table 3 were used to validate the simulation results, as demonstrated in Fig. 6. Specifically, the densification results of the numerical modelling were illustrated in the top plots in Fig. 6 for each case, respectively, which were compared with the corresponding experimental results below from Ref. [12]. Figure 6 illustrates that the distribution results for the densification degree from numerical calculation agree well with the experimental micrographs.

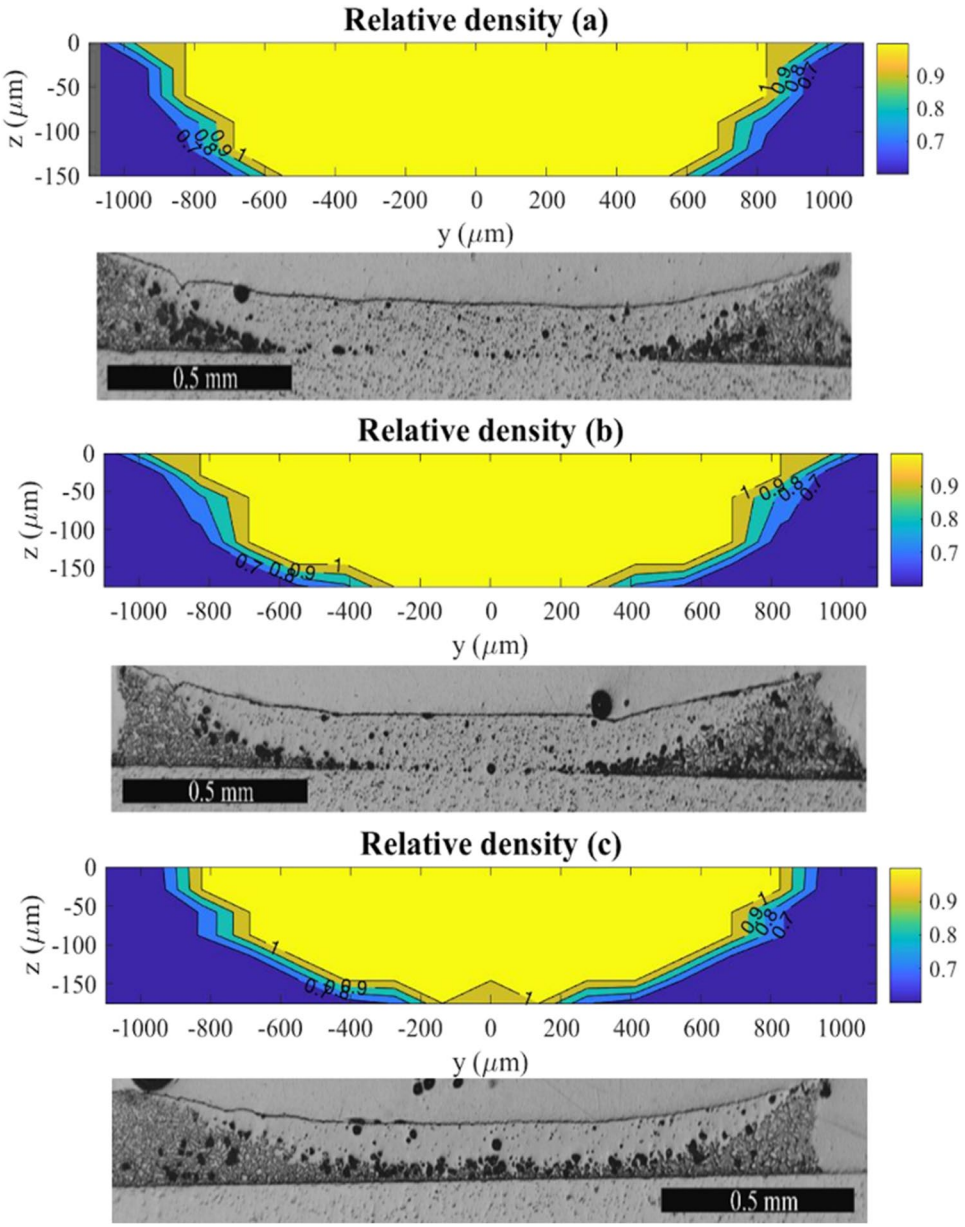


Fig. 6 The densification degree of the glass monolayer at the study plane and the experimental micrograph from Ref [12]

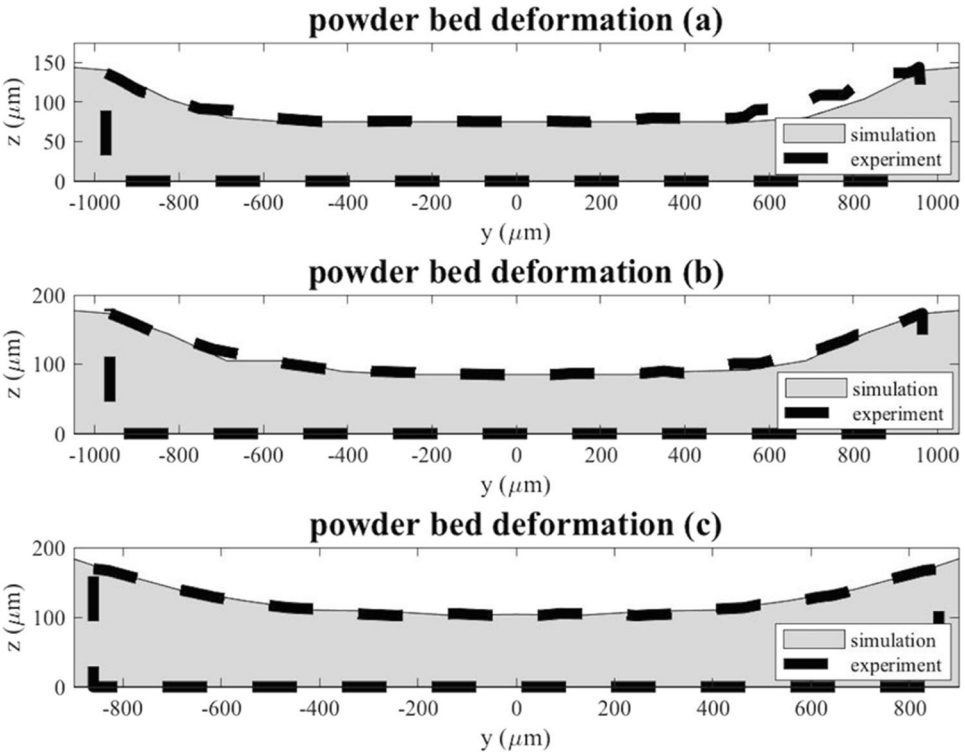


Fig. 7 Deformation comparison for the glass monolayer under SLM between the experiment in Ref [12] and the simulation in this work

The experimental micrographs presented in Fig. 6 from Ref [12] illustrate the lens-shaped domains of the powder bed in the cross-section perpendicular to the scanning direction. These domains have a concave shape, and the concavity of the top surface can be attributed to the uniform shrinkage of the powder during processing [12]. Particularly, the lighter-colour parts indicate the fused glass, while the darker-colour parts correspond to the nonconsolidated powder. Additionally, the experimental observation from Fig. 6 demonstrates the transition of the glass powder from an unconsolidated to a consolidated state after processing. The comparison in Fig. 6 validates that the numerical modelling can predict the transition trends in these three instances, which correspond to the shift of the relative density from 0.6 to 1.0.

The shrinkage deformation of the powder bed can be simulated based on the modelling results since the deformation was dependent on the different densification degrees. The linear relation was assumed for the density ρ_b and the height h of the powder bed.

$$h = h_m + \frac{h_s - h_m}{\rho_{bs} - \rho_{bm}}\left(\rho_b - \rho_{bm}\right) \tag{25}$$

where h_m, h_s, ρ_{bm} and ρ_{bs} denote the bed height of the middle and the side and the compact density on the corresponding position, respectively. The comparison between the simulation and the experimental results is shown in Fig. 7.

Table 4 Deformation comparison based on cross area between experiments and simulation

Case	Cross area from experiments (mm^2)	Cross area from simulation (mm^2)	Error
a	0.1181	0.1158	1.938%
b	0.1267	0.1264	0.197%
c	0.1485	0.1501	1.072%

The experimental deformation results were obtained by using ImageJ to capture the contour of the experiments demonstrated in Fig. 7 [12]. It was noted that the simulation could capture the bed deformation well in experiments. Table 4 lists the deformation comparison results based on the cross area between experiments and simulation. Particularly, the prediction errors for all the simulations of these three cases are less than 2%. Thus, it is concluded that this numerical modelling method can be used to simulate the SLM process of glass silica monolayer.

3.3 Discussion of the processing parameters

The effect of powder size was discussed by using this modelling method, in which the particle diameter decreased from 20 to 5 μm and other parameters were set as same as the case (a) in Table 3. The simulation results are illustrated in Fig. 8.

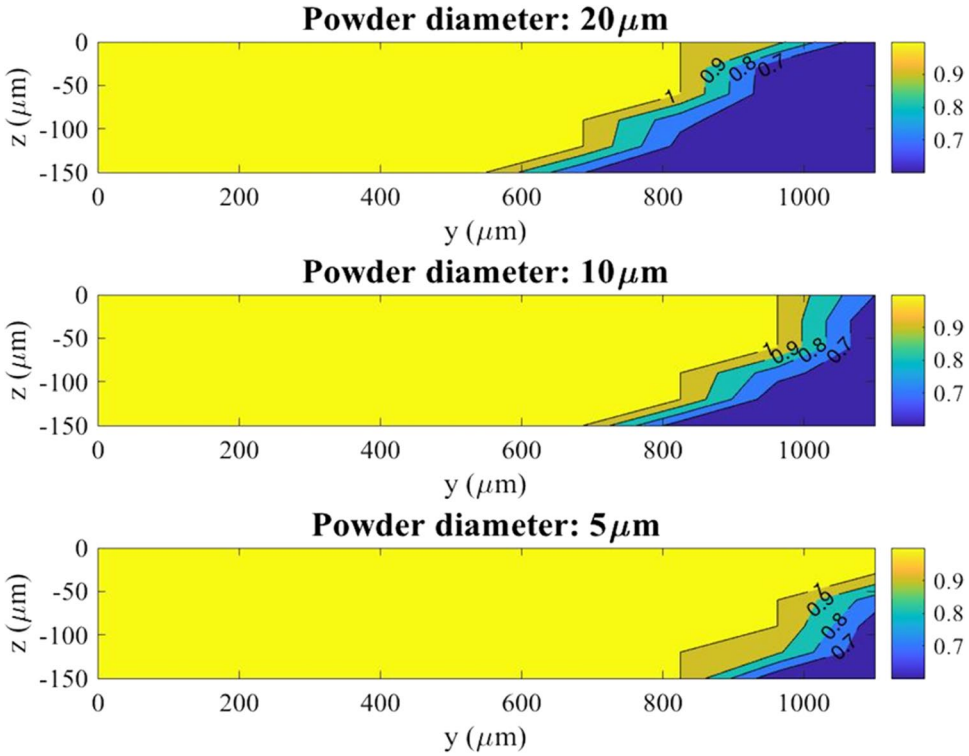


Fig. 8 Densification results of SLM for glass compacts with different powder diameters

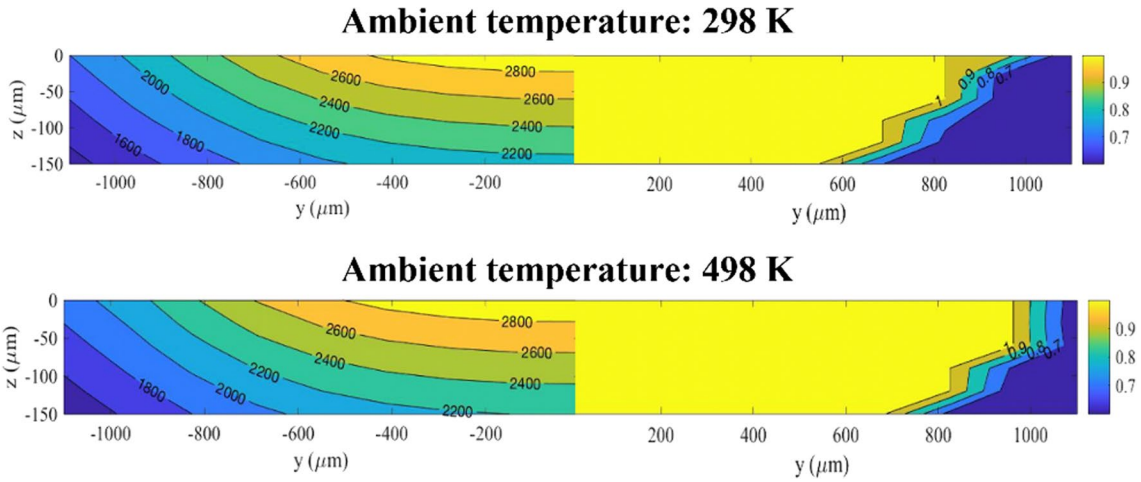


Fig. 9 The temperature distribution and densification degree of the glass monolater under SLM with different initial ambient temperature

Based on the necking kinetics theory [17] and the clusters model [14], the densification rate has an inverse relationship with the particle diameter. Besides, the experimental study also notes that the powder with a large diameter hinders the air bubbles from escaping and then traps them inside, which reduces the quality of the final part [26]. Therefore, the finer particles can speed up the consolidation process and improve the quality of the obtained parts. However, the influence of the glass powder size is not observed in the previous reference [12]. By contrast, it is noticeable in this model that the smaller particle diameter positively affects the process of SLM for silica glass, as demonstrated in Fig. 8.

Preheating is an effective technique to improve the quality of SLM, as demonstrated by the previous experiment [27, 28]. We set the initial ambient temperature from 298 to 498 K in our numerical model to discuss the effect of preheating on the SLM processing of glass. The simulation

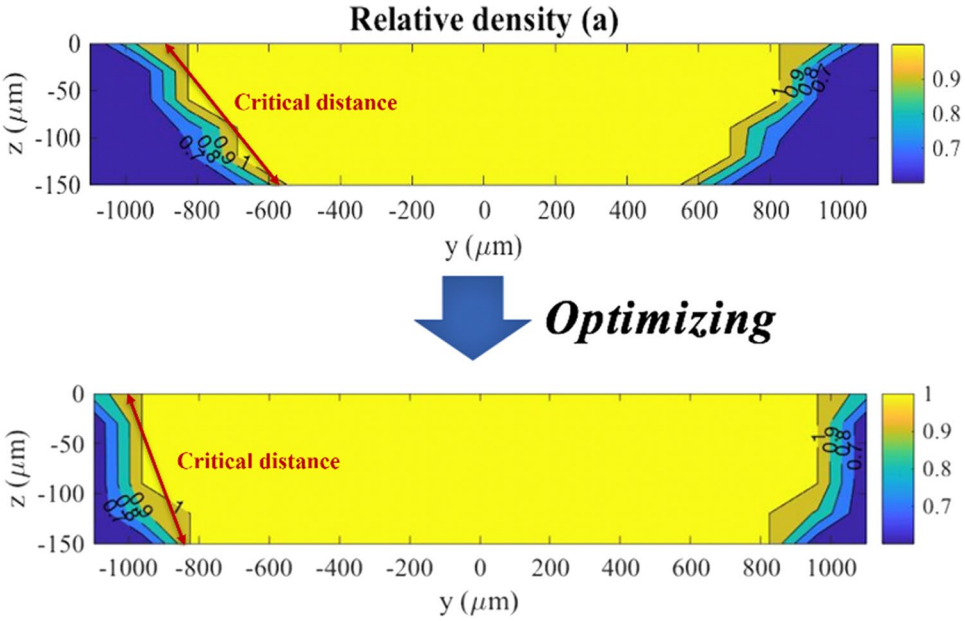


Fig. 10 An instance of the optimization criterion for the processing parameters of glass monolayer under SLM

results with other parameter settings same as case a are demonstrated in Fig. 9.

As depicted in Fig. 9, the thermal gradient is reduced at the initial ambient temperature. Figure 9 also illustrates that the densification process is accelerated for the glass monolayer under SLM. Thus, it is concluded that preheating can enhance the quality of the final glass part due to the lower thermal gradient and increase the manufacturing efficiency of SLM.

3.4 Processing optimisation

The densification results shown in this numerical study agreed well with the experimentally observed micrographs in Fig. 6. The dark yellow band in Fig. 6, representing the relative density from 1 to 0.95, could be matched with the boundary line between the fused glass and the non-consolidated glass. At the bottom of the monolayer in Fig. 6, the point between the consolidated glass and non-consolidated powders was defined as the critical point of the bottom. Meanwhile, this kind of point at the top surface was defined as the critical point of the top. As illustrated in Fig. 10, the distance between the two critical points was defined as the critical distance, which could work as an optimisation criterion. In other words, the closer the distance is, the better the SLM quality is.

Thus, a new optimisation scheme could be made according to this observation. The design variables were the parameters such as laser power, scanning velocity, layer thickness, particle size and ambient temperature. Minimising the critical distance was identified as the objective function. Additionally, the constraints included the maximum or minimum values of the variables, such as the range of the laser power, the required width scope of the fused glass and the maximum ambient temperature for preheating. We could take Fig. 10 as an example of the optimisation scheme established in this study. The densification results of case (a) were optimised by increasing ambient temperature from 298 to 498 K and decreasing particle size from 20 to 10 μm.

4 Conclusion

In this study, a numerical model was established to predict the temperature distribution, densification kinetics and deformation of the monolayer of glass powders under SLM. Specifically, the necking growth and densification kinetics were incorporated into the heat transfer model, which was calculated using FDM. The experimental observations from other research could validate the prediction results of the newly developed model with a prediction error of less than 2%. Thus, the well-proved numerical simulation could provide a more reliable tool to discuss the influence of the processing parameters. It was noted that the smaller size of the powders and the higher ambient temperature can speed up the consolidation process and improve the quality of the obtained parts. Besides, a new optimisation criterion based on the model can be set to optimise the parameters and fabricate the transparent and high-purity glass through SLM. In the future, researchers can use this numerical modelling method for glass monolayer to aid their 3D design and fabrication of fused silica parts by using SLM.

Author contribution Wanrui Zhang: conceptualization, methodology, formal analysis, investigation, writing.

Declarations

Competing interests The author declares no competing interests.

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